Neurodevelopmental Disorder With Movement Abnormalities, Abnormal Gait, And Autistic Features (NEDMAGA)

Causal gene: *ZSWIM6* (Zinc finger SWIM-type containing 6). NEDMAGA (OMIM **#617865**; MONDO:0060642) is an ultra-rare, autosomal-dominant neurodevelopmental disorder caused by a single recurrent **de novo** heterozygous nonsense variant, **NM_020928.2:c.2737C>T (p.Arg913Ter / R913X)**, in the penultimate exon (exon 13) of *ZSWIM6* on chromosome 5q12.1. It was delineated by Palmer et al. 2017 (PMID 29198722) in 7 unrelated individuals with severe-to-profound intellectual disability **without** the frontonasal/limb malformations seen in the allelic *ZSWIM6* disorder acromelic frontonasal dysostosis (AFND, OMIM #603671; caused by the different missense variant c.3487C>T, p.Arg1163Trp). This report follows the MONDO ID and the stated causal gene; the name correctly points to *ZSWIM6*.

> **Evidence base.** Information is aggregated from fewer than ~10 individual patients described in the primary literature (Palmer 2017, n=7, PMID 29198722 ; Yanagishita 2021, n=1 Japanese patient, PMID 33958584 ; Corazza 2024 case report, PMID 38386481) plus disease-level resources (OMIM #617865, ClinVar, UniProt, MGI, ZFIN) and model-organism studies (mouse: Tischfield 2017, PMID 28433741 ; zebrafish: Smith 2014, PMID 25105228).

> No large registry, EHR cohort, or natural-history study exists. Evidence source types are flagged throughout as **[human clinical]**, **[model organism]**, **[in vitro]**, or **[computational/database]**.

1. Disease Information

- **Overview.** NEDMAGA is characterized by infantile-onset global developmental delay, severe-to-profound intellectual disability, mildly delayed walking with a broad-based unsteady gait, absent/minimal language, autistic features with a characteristic *happy demeanor*, and a variety of movement abnormalities (stereotypies, spasticity, dyskinetic movements). It is a multisystem disorder with both central and peripheral nervous system involvement. **[human clinical]**
- **Key identifiers:**
- **OMIM:** #617865 (disease); *ZSWIM6* gene *615951
- **MONDO:** MONDO:0060642
- **ICD-11:** best mapped to LD90.Y / 6A00-range (disorders of intellectual development / developmental disorders); no dedicated code.
- **ICD-10:** no specific code (maps loosely to F79 / F84).
- **MeSH:** no dedicated term; covered by "Neurodevelopmental Disorders" (D065886).
- **Orphanet:** *ZSWIM6* listed by the Gene Curation Coalition; no dedicated ORPHA number with prevalence retrievable.
- **ClinVar:** Variation ID 431797 (rs1554041295); RCV000578491.
- **Synonyms / alternative names:** NEDMAGA; ZSWIM6-related intellectual disability without frontonasal or limb malformations; ZSWIM6-related neurodevelopmental disorder.

2. Etiology

- **Primary cause — genetic.** A single recurrent **de novo** heterozygous nonsense variant in *ZSWIM6*, c.2737C>T (p.Arg913Ter). All tested cases arose de novo (not inherited). **[human clinical]** Direct quote
(P 29198722):

"seven unrelated individuals with a recurrent de novo nonsense variant (c.2737C>T [p.Arg913Ter]) in the penultimate exon of ZSWIM6 who have severe-profound intellectual disability."

- **Genetic risk factors.** The causal variant is the sole known genetic determinant; no susceptibility loci or modifier genes have been identified. The variant is a C>T transition at a CpG-associated position (consistent with recurrent de novo mutability).
- **Environmental risk factors.** None identified. As a de novo monogenic disorder, the only recognized correlate of de novo mutation burden generally is advanced paternal age (not specifically demonstrated for NEDMAGA). **Not established for this disease.**
- **Protective factors (genetic or environmental).** None known. **Not applicable.**
- **Gene–environment interactions.** None described. **Not applicable** (fully penetrant monogenic etiology).

3. Phenotypes

Frequencies below are from the 7-patient Palmer 2017 cohort unless noted. Because n is tiny, frequencies are approximate.

Phenotype	Type	Onset	Severity	Course	Frequency	HPO suggestion
Global developmental delay	sign	infantile/ congenital	severe	static→slowly progressive	~100%	HP:000126
Intellectual disability, severe–profound	sign	infantile	severe-profound	static	~100%	HP:001086 HP:000218
Absent or minimal speech	sign	childhood	severe	static	~100%	HP:000134
Autism / autistic features	behavioral	early childhood	variable	static	5/7 (~71%)	HP:000071 HP:000072
Hyperactivity	behavioral	childhood	mild-mod	static	4/7 (~57%)	HP:000075
Happy demeanor / sociable	behavioral	childhood	—	static	~100%	HP:010002 (cheerful disposition
Broad-based, unsteady gait / ataxic gait	sign	toddler (walk 2–5 yr)	mod-severe	some lose ambulation	high	HP:000213 HP:000231
Hypotonia	sign	infantile	mod	may evolve to hypertonia	high	HP:000125
Spasticity / hypertonia	sign	childhood	mod	progressive	3/7	HP:000125
Stereotypic hand movements	behavioral	childhood	—	static	present	HP:001217
Paroxysmal tongue movements / head tics	sign	childhood	—	episodic	present	HP:010002 (abnormal movement
Seizures (well-controlled)	sign	childhood	mild	episodic, treatable	3/7 (+1 suspected)	HP:000125

Phenotype	Type	Onset	Severity	Course	Frequency	HPO suggestion
Microcephaly (progressive)	physical	postnatal	—	progressive	3/7	HP:000025
Macrocephaly	physical	—	—	—	1/7	HP:000025
Coarse facies / prominent forehead / short nose / depressed nasal bridge	physical	progressive w/ age	mild	progressive	subset	HP:000028 HP:001122 HP:000319 HP:000528
Peripheral neuropathy (axonal + demyelinating)	sign/lab	childhood	mod	—	1/7 (biopsy-proven)	HP:000983
Hyporeflexia	sign	childhood	—	—	subset	HP:000126
Reduced pain sensitivity (→ self-injury)	sign	childhood	—	static	2/7	HP:000702
Failure to thrive / poor feeding	sign	infantile	mod	—	subset	HP:000150 HP:001196
Gastroesophageal reflux / vomiting	sign	infantile	variable→severe	may be refractory	subset	HP:000202 HP:000201
Constipation	sign	childhood	mild	chronic	subset	HP:000201

- **Quality-of-life impact.** Profound: non-verbal, dependent for most ADLs, mobility impaired (some non-ambulatory), lifelong care required. Formal QoL instruments (EQ-5D/SF-36/PROMIS) have not been applied. The *happy demeanor* and social responsiveness are notable positive features for caregiver interaction. **[human clinical]**

4. Genetic / Molecular Information

- **Causal gene:** *ZSWIM6* — HGNC:29316; NCBI Gene 57688; Ensembl ENSG00000130449; UniProt **Q9HCJ5** (ZSWM6_HUMAN); OMIM gene *615951; locus 5q12.1 (chr5:61,332,258–61,546,172, GRCh38). Paralog:* ZSWIM5*.
- **Pathogenic variant:** NM_020928.2:**c.2737C>T (p.Arg913Ter)**; dbSNP rs1554041295; ClinVar Variation 431797.
- **Classification: Pathogenic** (ACMG support: PS2 de novo, PS4 recurrent in affected/absent in controls, PM2 absent from gnomAD; caveat on PVS1 because it escapes NMD).
- **Type/class:** nonsense (stop-gain) single-nucleotide variant.
- **Allele frequency:** 0 — absent from gnomAD, 1000 Genomes, ESP. **[database]**
- **Origin:** germline, de novo (not somatic/mosaic in probands, though parental germline mosaicism cannot be formally excluded).
- **Functional consequence:** The variant lies at the 3′ terminus and **does not trigger nonsense-mediated decay**, producing a truncated protein lacking the C-terminal **Sin3-like domain** — proposed **dominant-negative** mechanism. Direct quote (P 29198722): *"This finding supports the existence of a truncated ZSWIM6 protein lacking the Sin3-like domain, which could have a dominant-negative effect."* **[in vitro, patient-derived cells]**
- **Allelic contrast:** AFND is caused by missense c.3487C>T (p.Arg1163Trp) in the same Sin3-like domain, proposed **gain-of-function** (P 25105228). Same gene → distinct phenotypes depending on variant.
- **Modifier genes:** none identified.
- **Epigenetic information:** no disease-specific methylation/histone signature reported; note the Sin3-like domain implies a role in transcriptional co-repressor (Sin3/HDAC) complexes, i.e., chromatin regulation, but this is inferred. **Not demonstrated for the disease.**
- **Chromosomal abnormalities:** none — a point mutation, not a CNV/structural change.

5. Environmental Information

- **Environmental factors, lifestyle factors, infectious agents: None — not applicable.** NEDMAGA is a monogenic de novo disorder with no known toxic, nutritional, behavioral, or infectious contribution.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. A **de novo** C>T mutation creates *ZSWIM6* c.2737C>T (p.Arg913Ter) **→ leads to** a premature stop codon in the penultimate exon. **[human clinical]**
2. Because the stop is 3′-terminal, it **escapes nonsense-mediated decay → results in** a stable, truncated ZSWIM6 protein lacking the C-terminal **Sin3-like domain**. **[in vitro, demonstrated — mRNA persists in patient cells]**
3. The truncated protein **is inferred to act as a dominant-negative** (interfering with wild-type ZSWIM6 and/or its partner complexes) rather than a simple loss-of-function. **[inferred]**
4. Loss/perturbation of normal ZSWIM6 function **→ dysregulates** its molecular activities: (a) a putative **E3 ubiquitin-ligase** function (ZSWIM6 ubiquitinates NAT10; P 38246918) and (b) a **Sin3-like transcriptional co-regulator** role, and **→ is linked to Hedgehog (SHH) pathway dysregulation** (the AFND allele activates Hedgehog; P 25105228). **[in vitro; branch — Hedgehog shown for the AFND allele, inferred for NEDMAGA]**
5. In the developing and postnatal brain, ZSWIM6 is **highly expressed in the striatum** (restricted postnatally), so its dysfunction **→ impairs striatal development**, reducing striatal volume and altering **medium spiny neuron (MSN)** morphology. **[model organism — mouse KO,** P 28433741 **]**
6. Abnormal striatal/MSN circuitry **→ produces** dysregulated motor control: hyperactivity, impaired motor coordination, repetitive/stereotypic movements, and altered dopaminergic responsiveness (amphetamine hyperresponsiveness in mice). **[model organism]** In

humans this → **manifests as** broad-based unsteady gait, stereotypies, hyperkinetic movements, and autistic repetitive behavior. **[human clinical, inferred linkage]**

7. In parallel, ZSWIM6 dysfunction in peripheral nerve → **causes** a mixed axonal/demyelinating neuropathy (reduced myelinated-fiber density, thin myelin, abnormal neurofilament accumulation on sural-nerve biopsy) → **producing** hyporeflexia and reduced pain sensitivity. **[human clinical, 1 biopsy-proven case]**
8. Broad CNS developmental disruption → **results in** severe-profound intellectual disability, absent language, microcephaly, and (in some) seizures. **[human clinical]**

Supporting detail by category: - **Molecular pathways:** Hedgehog/SHH signaling (GO:0007224 smoothened signaling) — activated by AFND allele, inferred relevance here; ubiquitin–proteasome pathway (via ZSWIM6 E3-ligase activity on NAT10); Sin3/HDAC transcriptional co-repression (inferred from the Sin3-like domain). - **Cellular processes:** neuronal development and differentiation (GO:0030182), striatal MSN morphogenesis, regulation of locomotion/motor behavior (GO:0040012), protein ubiquitination (GO:0016567), myelination (GO:0042552, peripheral). - **Protein dysfunction:** truncation removing the Sin3-like domain → dominant-negative; contrast gain-of-function missense in AFND. - **Molecular profiling:** qRT-PCR of AFND patient osteoblast/fibroblast lines suggested Hedgehog activation (P 25105228) **[in vitro]**; no transcriptomic/proteomic/metabolomic/lipidomic dataset exists specifically for NEDMAGA. - **Functional genomics:** *Zswim6* knockout mouse (reverse genetics) is the principal functional model. No CRISPR/RNAi screens reported. - **Immune/metabolic involvement:** none described. **Not applicable.** - **Cell types (CL):** medium spiny neuron (CL:1001474); striatal neuron; myelinating Schwann cell (CL:0002573, peripheral neuropathy). - **GO biological processes to annotate:** GO:0007399 (nervous system development), GO:0021799 (cerebral cortex development), GO:0007224 (Hedgehog), GO:0016567 (protein ubiquitination).

7. Anatomical Structures Affected

- **Organ level / body systems:** Primarily the **central nervous system** (brain) and **peripheral nervous system**; secondary involvement of the **gastrointestinal system** (GERD, constipation, feeding) and **musculoskeletal/muscle** (tone abnormalities). Craniofacial structures mildly affected (dysmorphism). **[human clinical]**
- **Specific CNS sites (UBERON): striatum** (UBERON:0002435) and caudate/putamen (UBERON:0001873/UBERON:0001874) — principal site per mouse model; cerebral cortex

(UBERON:0000956, mild atrophy in 1 patient); telencephalon (UBERON:0001893); midbrain/hindbrain (developmental expression). Peripheral nerve (UBERON:0001021); sural nerve (UBERON:0003716).

- **Tissue/cell level:** nervous tissue; medium spiny neurons (CL:1001474); peripheral myelinated axons / Schwann cells (CL:0002573).
- **Subcellular (GO cellular component):** nucleus (GO:0005634, for Sin3-like transcriptional function); cytoplasm; the neuropathy shows abnormal **neurofilament** (cytoskeletal) accumulation. **[human clinical, biopsy]**
- **Lateralization:** bilateral / symmetric (generalized neurodevelopmental involvement).

8. Temporal Development

- **Onset:** congenital/infantile; global developmental delay evident in infancy; walking delayed to 2–5 years. **Onset pattern:** chronic/insidious (developmental). **[human clinical]**
- **Progression:** largely static developmental disability, but with some **progressive** features — progressive spasticity (3/7), progressive microcephaly (3/7), progressively apparent facial dysmorphism with age, and possible loss of ambulation. Overall slow/variable. **Disease duration:** chronic, lifelong.
- **Patterns:** no remission; seizures are episodic but well-controlled. **Critical period:** early neurodevelopment (prenatal–early childhood) — the window where striatal/cortical circuitry is established; no proven intervention window exists.

9. Inheritance and Population

- **Epidemiology:** Ultra-rare; ~8–10 reported individuals worldwide. Palmer et al. estimated the frequency of *ZSWIM6* pathogenic variants at **1.7–4.9 × 10^{-4}** among patients with neurocognitive disorders (P 29198722). True population prevalence/incidence undefined (likely <1/1,000,000). **[human clinical]**
- **Inheritance: Autosomal dominant**, essentially always **de novo**. **Penetrance:** complete in reported cases. **Expressivity:** relatively consistent core phenotype with variability in seizures, head size, neuropathy, and GI severity.

- **Genetic anticipation:** not applicable (not a repeat-expansion disorder). **Germline mosaicism:** not reported but theoretically possible → small recurrence risk for apparently unaffected parents. **Founder effect:** none (recurrent de novo at a mutational hotspot, not inherited). **Consanguinity:** not relevant (dominant de novo). **Carrier frequency:** not applicable (variant absent from population databases).
- **Population demographics:** Reported across diverse populations (European-ancestry cohorts in Palmer 2017; a Japanese patient in Yanagishita 2021), indicating no ethnic restriction. **Sex ratio:** not established (small n; includes males and females). **Age distribution:** children through adults (reported ages 3–29 years).

10. Diagnostics

- **Primary diagnostic test — genetic.** Diagnosis is molecular: identification of the de novo **c.2737C>T (p.Arg913Ter)** variant in *ZSWIM6*.
- **WES / WGS:** first-line and the method by which the disorder was discovered; trio (proband + parents) exome/genome sequencing demonstrates de novo status. High utility.
- **Gene panels:** intellectual disability / autism / epilepsy NGS panels that include *ZSWIM6*.
- **Single-gene / targeted testing:** Sanger confirmation of the recurrent variant.
- **CMA / karyotype / FISH:** normal (no structural abnormality) — useful to exclude differentials, not diagnostic here.
- **mtDNA / repeat-expansion testing:** not applicable.
- **Supportive clinical tests (characterize, not diagnose):**
- **Brain MRI:** usually **normal** (5/7); mild cortical atrophy in 1. **[human clinical]**
- **EEG:** for seizure evaluation in affected subset.
- **Nerve conduction studies / EMG:** may show mixed axonal–demyelinating neuropathy.
- **Sural nerve biopsy:** reduced myelinated-fiber density, thin myelin, neurofilament accumulation (research/selected cases).
- **Routine labs/metabolic workup:** non-specific/normal; no diagnostic biomarker exists.
- **Biomarkers:** none; the genetic variant is the only specific marker. No FDA biomarker.
- **Clinical criteria / differential diagnosis:** No formal criteria. Differentials include Rett syndrome and *MEF2C*-related disorder (NEDHSIL, OMIM #613443 — overlapping stereotypies/hypotonia/absent speech), Angelman syndrome (happy demeanor, ataxic gait, ID), other ID/ASD syndromes, and hereditary spastic paraplegia/CMT-overlap (for the

neuropathy/spasticity). *ZSWIM6*-AFND is distinguished by frontonasal/limb malformations (absent in NEDMAGA). Resolution is by genetic testing.

- **Screening:** No newborn/population screening. Cascade testing of parents is for recurrence-risk counseling; because variant is de novo, parental testing is typically negative.

11. Outcome / Prognosis

- **Survival/mortality:** No systematic mortality data; patients survive into adulthood (oldest reported 29 years). Life-threatening risk relates to severe refractory GERD/aspiration and feeding failure in the most severe cases. **[human clinical]**
- **Morbidity/disability:** Severe lifelong disability — non-verbal, intellectual disability, impaired mobility (some non-ambulatory), dependence for ADLs. ICF domains broadly affected.
- **QoL measures:** none formally applied.
- **Complications:** seizures, spasticity/contractures, feeding difficulties/GERD with vomiting (can be refractory even after gastrostomy), self-injury from reduced pain sensitivity, aspiration.
- **Recovery potential:** none for the core disability; supportive care improves function/comfort.
- **Prognostic factors:** severity of feeding/GI involvement and ambulation status are the main functional determinants; no molecular prognostic biomarker (single recurrent variant).

12. Treatment

No curative or disease-specific/targeted therapy exists; management is entirely supportive and symptom-directed.

- **Pharmacotherapy (symptomatic):**
- **Antiepileptic/anticonvulsant drugs** for seizures (well-controlled in reported patients). NCIT: Anticonvulsant Agent.
- **Anti-reflux therapy** (proton-pump inhibitors, prokinetics) for GERD/vomiting; NCIT: Proton Pump Inhibitor.
- **Laxatives/bowel regimen** for constipation.

- Agents for spasticity (e.g., baclofen) and behavioral/hyperactivity management as clinically indicated.
- **Pharmacogenomics:** none specific to *ZSWIM6*.
- **Advanced therapeutics (gene/cell/RNA/targeted/immuno): None available or in trials.** Conceptually, a dominant-negative, NMD-escaping allele might be amenable to allele-selective antisense-oligonucleotide/RNA knockdown, but this is hypothetical and not under development. **Not applicable currently.**
- **Surgical/interventional: Gastrostomy (PEG)** for feeding failure (note GERD may persist post-PEG; anti-reflux surgery/fundoplication may be considered). NCIT: Gastrostomy.
- **Supportive/rehabilitative: Physical therapy** (gait, spasticity, contracture prevention), **occupational therapy**, **speech/communication therapy** (AAC given absent language), nutritional support, orthopedic/management of tone, protective measures for pain-insensitivity self-injury. NCIT: Physical Therapy; Occupational Therapy; Speech Therapy; Supportive Care.
- **Experimental trials:** none (no NCT identifiers).
- **Treatment outcomes/strategy:** symptom-based multidisciplinary care; seizures generally respond well; GERD may be refractory. No treatment algorithm or precision-medicine approach established.

13. Prevention

- **Primary prevention:** Not possible for a sporadic de novo disorder. **Genetic counseling** is central: recurrence risk for parents of an affected child is low (de novo), but ~**50%** for an affected individual's own offspring (dominant) — though reproduction is unlikely given severity.
- **Secondary prevention (early detection):** Early genetic diagnosis via trio WES/WGS enables early intervention (developmental therapies) and anticipatory management of seizures/GI issues. **Prenatal testing / preimplantation genetic testing (PGT)** are available for the known variant in the rare situation of a prior affected child (covers germline-mosaicism risk).
- **Tertiary prevention:** Prevent complications — aspiration/reflux management, seizure control, contracture prevention (PT), protection from self-injury, nutritional optimization.
- **Immunization / behavioral / public-health / environmental interventions:** Not applicable (no environmental/infectious component).

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** *ZSWIM6* is conserved across vertebrates — human (NCBI Taxon 9606), mouse *Zswim6* (Taxon 10090; NCBI Gene 67263; MGI:1914513; ENSMUSG00000032846), rat *Zswim6* (Taxon 10116; NCBI Gene 310062; RGD 1306624), zebrafish *zswim6* (Taxon 7955; ZFIN ZDB-GENE-060602-4). 1:1 orthology (stringent).
- **Natural disease in other species: None reported** — no entry in OMIA; no naturally occurring NEDMAGA-equivalent in companion animals or wildlife. No breed (VBO) associations. **Not applicable.**
- **Comparative biology:** Conserved brain expression (striatal/telencephalic) across mouse and zebrafish supports an evolutionarily conserved neurodevelopmental role.
- **Transmission / zoonosis:** Not applicable (genetic, non-communicable).

15. Model Organisms

- **Mouse — *Zswim6* knockout (Tischfield et al. 2017,**

  P 28433741

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  [model organism]. Mammalian, germline knockout. **Phenotype recapitulation:** reduced striatal volume, altered medium-spiny-neuron morphology, hyperactivity, impaired rotarod performance, repetitive movements, amphetamine hyperresponsiveness — recapitulating NEDMAGA's motor-regulation and repetitive/autistic-like features and localizing function to the striatum. Direct quote: *"Loss of Zswim6 causes a reduction in striatal volume and changes in medium spiny neuron morphology. These changes are associated with alterations in motor control, including hyperactivity, impaired rotarod performance, repetitive movements, and behavioral hyperresponsiveness to amphetamine."* **Limitation:** it is a loss-of-function model, whereas the human allele is a NMD-escaping, truncating, proposed **dominant-negative** variant; the KO may therefore under- or mis-represent human pathobiology; craniofacial/AFND features and the specific human truncation are not modeled. Resource: MGI:1914513; IMPC.
- **Zebrafish — *zswim6* (Smith et al. 2014,**

  P 25105228

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  [model organism]. Used for developmental expression mapping (telencephalic at 24 hpf; midbrain/hindbrain/retinal at 48 hpf), supporting a neurodevelopmental role. Resource: ZFIN ZDB-GENE-060602-4.

- **In vitro / cellular models:** patient-derived osteoblast/fibroblast lines (AFND, Hedgehog qRT-PCR; P 25105228) and patient-derived cells showing NMD escape of the NEDMAGA allele (P 29198722). No iPSC/organoid model reported for NEDMAGA.
- **Applications:** mouse KO enables study of striatal circuitry, motor behavior, and MSN biology; zebrafish for developmental expression. **Gaps:** no knock-in model of p.Arg913Ter to test the dominant-negative hypothesis.

Limitations and Future Directions

- **Evidence is extremely sparse** (<10 patients, single defining cohort), so frequencies and genotype–phenotype correlations are preliminary; sex ratio, prevalence, and natural history are undefined.
- **Mechanism is partly inferred:** the dominant-negative effect and Hedgehog involvement for the NEDMAGA allele are hypotheses extrapolated from the AFND allele and the Sin3-like domain; NMD escape and striatal expression are the best-demonstrated steps.
- **No targeted therapy or trials;** an allele-selective knockdown strategy is a logical but untested future direction.
- **Priority future work:** a knock-in p.Arg913Ter mouse/iPSC-neuron model to test dominant-negative vs loss-of-function; transcriptomic/proteomic profiling of patient neurons; a patient registry (humandiseasegenes.nl/zswim6) to define natural history; clarification of the severe refractory GERD mechanism.

Key references

- Palmer EE, et al. *Am J Hum Genet.* 2017. P 29198722 — defining NEDMAGA cohort, p.Arg913Ter, NMD escape, dominant-negative hypothesis. [human clinical / in vitro]

- Smith JD, et al. *Am J Hum Genet.* 2014.

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  — AFND, p.Arg1163Trp, Hedgehog activation, zebrafish expression. [human clinical / model organism]

- Tischfield DJ, et al. *Neurobiol Dis.* 2017.

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  — *Zswim6* knockout mouse striatal/motor phenotype. [model organism]

- Yanagishita T, et al. *Hum Genome Var.* 2021.

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  — second NEDMAGA patient; epilepsy, microcephaly, refractory GERD. [human clinical]

- Corazza G, et al. *Mov Disord Clin Pract.* 2024.

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  — NEDMAGA case report (movement-disorder perspective). [human clinical]

- Wang Y, et al. 2024.

  P 38246918

  — ZSWIM6 as an E3 ubiquitin ligase (NAT10). [in vitro]